

Super-Lee Management System - User Guide:

1. Overview

The Super-Lee Management System is a console-based application used to manage employees, roles, and shift assignments in a supermarket. It also supports driver management, delivery-related validations, and SQLite database persistence.

The system supports two types of users:

- HR Manager
- Employee

2. Running the System

To run the system:

1. Open the project in IntelliJ / VS Code
2. Navigate the file:
presentation/Main.java
3. Run the main method:
 - On startup, the system connects to the SQLite database file superLee.db. If the file does not exist, it is created automatically.
 - After The connection is established, the system starts and displays the main menu.

Startup database behavior:

- The submitted project includes an initial SQLite database file named superLee.db with sample data.
- When the program starts, it checks whether the employees' table is empty.
- If the database is empty, initial mock data is loaded automatically.
- If the database already contains data, the system does not load mock data automatically, in order to avoid mixing or damaging user-created data.

3. Main Menu

Welcome to Super-Lee management system

Select user profile:

1. HR Manager
2. Employee
3. Load mock data
4. Reset system (Delete all data)
4. Exit

HR Manager - access to management features

Employee - access employee features

Load mock data- Loads predefined sample data only when the database is empty. If data already exists, the load is blocked or skipped to avoid mixing data.

Reset system (Delete all data)- Deletes all data from all tables after confirmation.

Exit - close the system

4. User Identification

User Identification:

The system supports two types of users:

- HR Manager:** requires a password to access management features

-**Employee:** identified using Employee ID only (no password required)

*When selecting HR Manager, the user must enter the correct password

*When selecting Employee, the user is required to enter a valid Employee ID

The HR Manager password is: aadoss

5. HR Manager Capabilities

After selecting HR Manager, the following menu appears:

1. Employee management
2. Shift & assignment management
3. Back

6. Employee Management

This section allows full management of employees.

Available actions:

1. Add new employee
2. Update employee details
3. View employees details
4. Fire employee

-Adding a New Employee:

The system will request:

- Name
- Id
- Bank account number
- Employment type: Full-time\Part-time
- Salary type: Hourly\Global
- Salary amount
- Vacation days
- The manager is asked whether the employee is a driver. If yes, the system also requests license type.

Notes:

- Invalid input will be rejected
- Employee IDs must be unique

-Managing Roles:

You can:

- Add a role to an employee
- Remove a role from an employee

The system prevents duplicate roles (case-insensitive).

-Firing an Employee

Fired employees are:

- Not deleted
- Marked as inactive

Their history remains stored in the system.

*Current predefined roles include Cashier, Warehouse, and Driver. Additional store roles can be added by the HR Manager.

7.Shift & Assignment Management

This section allows management shifts and assignments:

1. Shift schedule (create, view history)
2. Assignment (assign, edit, remove, replace manager)
3. System settings (closed days, weekly deadline)
4. View employee availability
5. Add new store role
6. Validate delivery shift requirements

Available actions:

- Create new shift
- View shift history
- Assign employees to shift
- Replace shift manager
- Remove employees from shift
- Edit assignment
- View employee availability
- Set weekly deadline
- Add closed day
- Add new store role
- Validate delivery shift requirements

-Creating a Shift:

To create a shift:

- 1.Enter day
- 2.Enter shift type:
 - Morning
 - Evening
 - Double
3. Define required roles and number of employees

-Selecting a Shift Manager

Before assigning employees, a shift manager must be selected.

Steps:

- 1.Choose "Replace shift manager"
- 2.Enter shift details
- 3.Enter employee ID

The employee must:

- Exist
- Be active
- Be certified as a shift manager

Assignment is not allowed without a shift manager!

-Assigning Employees to Shifts

To assign an employee:

- 1.Enter shift day
- 2.Enter shift type
- 3.Enter employee ID
- 4.Select role

Conditions:

- Employee must exist and be active
- Employee must have the required role
- Shift must exist
- Shift must not be full for that role
- Employee must not already be assigned

-Assignment Outside Availability

If the employee is not available for the shift:

The system will display a warning

The manager may choose to override the restriction

-Removing an Employee from a Shift

You can remove an employee by:

Employee ID

Day

Shift type

If the removed employee was the shift manager, the shift will no longer have a manager.

-Editing Assignments

You can:

Change an employee's role in a shift

Move an employee between shifts

-Viewing Shift History

Displays:

Day and shift type

Shift manager

Assigned employees

Each employee's role and ID

-Viewing Employee Availability

Allows the manager to see availability submitted by employees.

-Weekly Deadline

The HR Manager can set a deadline for availability submission.

After the deadline:

Employees cannot add availability

Employees cannot update availability

Employees cannot delete availability

-Closed Days

The manager can define closed days.

On closed days:

No shifts can be created

No availability can be submitted

-Delivery Shift Validation

-The delivery module is represented by DeliveryServiceMock.

-The current mock defines a delivery on Sunday morning.

- The required truck license for that delivery is B.
- The assigned delivery driver in the mock is employee 1010.
- The validation checks: a delivery exists for the shift, a driver is assigned to the delivery, the driver appears in the shift assignment schedule, the driver license matches the truck requirement, and a Warehouse employee is assigned in the same shift.

8.Employee Capabilities

After selecting **Employee**, the user must enter their ID.
If valid, the employee menu appears.

-Available Actions

- Manage availability
- View personal shifts
- View shift board
- Access shift manager functions (if certified)

-Managing Availability

Employees can:

- Add availability
- Delete availability
- Edit availability
- View availability
- Availability is defined by:

Day

Shift type (Morning / Evening / Double)

Important Note:

If an employee does not submit availability at all:
The system assumes the employee is available for all shifts

-Viewing Shifts

Employees can see all shifts they are assigned to.

-Shift Manager Menu

If the employee is certified as a shift manager, additional options are available.

9.Business Rules

- A shift must have a shift manager before assigning employees
- Employees can work multiple shifts in one day
- Employees cannot be assigned to both "Double" and another shift in the same day
- Employees without submitted availability are considered available for all shifts.
- Fired employees remain in the system but are inactive
- Closed days block shifts and availability

10.Initial Data and Mock Data

- The project is submitted with superLee.db containing initial sample data.
- The system also includes initMockData in Main.java to restore sample data if the database is empty.
- At program startup, if the database is empty, the system loads the initial mock data automatically.
- If the database already contains data, mock data is not loaded. This prevents mixing sample data.
- The manual Load mock data option follows the same idea: it is intended for an empty database.

-The Reset system option clears all tables after confirmation. After reset, mock data can be loaded again automatically on the next run or manually from the menu.

-Important Reset Warning:

Reset system is a hard reset. It permanently deletes all records from the module database tables: employees, employee_roles, employee_availabilities, shifts, shift_requirements, and shift_assignments. Use it only when you intentionally want a clean database for testing.

11.Database Structure

The project uses SQLite.

Package/Class	Responsibility
dataAccess.Database.DataBase	Creates the SQLite connection to superLee.db and creates the tables if they do not exist. Also includes clearAllTables for reset.
dataAccess.DAO.EmployeeDAOImpl	Inserts, updates, and loads employees, their roles, and their availabilities.
dataAccess.DAO.ShiftDAOImpl	Inserts, updates, and loads shifts and role requirements.
dataAccess.DAO.ShiftAssignmentDAOImpl	Inserts, updates, deletes, and loads shift assignments.
dataAccess.DTO.EmployeeDTO	Data transfer object for employees, including roles and availabilities.
dataAccess.DTO.ShiftDTO	Data transfer object for shifts and shift requirements.
dataAccess.DTO.ShiftAssignmentDTO	Data transfer object for assignments between employee, shift, and role.

Database Tables in the Initial DB:

Table	Main columns	What it stores	Initial sample content
employees	id, name, bank_account, active, is_manager, branch_id, employment_type, salary_type, salary, start_date, vacation_days, license_type	All employees. active is stored as 1/0. is_manager is stored as 1/0. Driver license is stored in license_type.	Sample employees include shift managers, cashiers, warehouse workers, and driver 1010 with license B.
employee_roles	employee_id, role_name	Role list for each employee.	Roles include Cashier, Warehouse, and Driver. Several employees share the same role so the tester can choose among alternatives.

employee_availabilities	employee_id, day, shift_type	Weekly availability submitted by employees.	Employees have different availabilities, enabling tests for valid assignment, unavailable assignment, and override.
shifts	day, shift_type, manager_id, branch_id	Defined shifts and selected shift manager for each shift.	Nine sample shifts are included: Sunday morning, Monday morning/evening, Tuesday morning/evening, Wednesday morning, Thursday morning, Friday morning/evening.
shift_requirements	day, shift_type, role_name, amount	Required number of workers per role for each shift.	Requirements include Cashier, Warehouse, and Driver. Sunday morning requires Driver = 1.
shift_assignments	employee_id, day, shift_type, role_name	Actual employee assignments to shifts.	Sample assignments include managers, cashiers, warehouse employees, and driver 1010 assigned to Sunday morning.

Initial Employees:

ID	Name	Type	Roles	License
111	Michael Brown	Shift Manager	Cashier	-
222	Bob Hill	Employee	Warehouse	-
333	Lola White	Shift Manager	Warehouse, Cashier	-
444	Diana Walker	Employee	Cashier	-
555	Ben Smith	Shift Manager	Warehouse	-
666	Emily Sue	Employee	Warehouse	-
777	Mia Wilson	Employee	Cashier	-
888	Johny Lee	Employee	Cashier	-
999	New worker	Employee	Warehouse	-
1010	Dan Driver	Driver	Driver	B

Initial Shifts and Requirements:

Shift	Manager	Requirements
Sunday morning	111	Cashier 2, Warehouse 2, Driver 1
Monday morning	333	Warehouse 1, Cashier 2
Monday evening	333	Warehouse 2, Cashier 2
Tuesday morning	111	Cashier 3, Warehouse 1
Tuesday evening	555	Cashier 2, Warehouse 1
Wednesday morning	111	Cashier 2, Warehouse 3
Thursday morning	555	Warehouse 2, Cashier 1
Friday morning	333	Warehouse 2, Cashier 2
Friday evening	333	Warehouse 2, Cashier 1

Initial Assignments:

Shift	Assignments
Sunday morning	111-Cashier (shift manager), 333-Warehouse, 777-Cashier, 555-Warehouse, 1010-Driver
Monday morning	333-Warehouse (shift manager), 111-Cashier, 444-Cashier
Monday evening	333-Warehouse (shift manager), 111-Cashier, 444-Cashier, 222-Warehouse
Tuesday morning	111-Cashier (shift manager), 444-Cashier, 888-Cashier
Tuesday evening	555-Warehouse (shift manager), 111-Cashier, 444-Cashier
Wednesday morning	111-Cashier (shift manager), 444-Cashier, 555-Warehouse, 666-Warehouse, 999-Warehouse
Thursday morning	555-Warehouse (shift manager), 444-Cashier, 222-Warehouse
Friday morning	333-Warehouse (shift manager), 444-Cashier, 888-Cashier, 555-Warehouse
Friday evening	333-Warehouse (shift manager), 222-Warehouse, 888-Cashier

Delivery Test Scenario in the Initial DB:

Item	Value
Delivery shift	Sunday morning
Mock driver ID	1010
Driver name	Dan Driver
Driver license	B
Required truck license	B
Warehouse requirement	Sunday morning includes Warehouse workers assigned to the shift
Expected validation result	Delivery shift requirements are satisfied.

Viewing the Database:

1. Open DB Browser for SQLite or an SQLite Viewer extension in VS Code.
2. Open the file superLee.db from the project root.
3. Use Browse Data to select tables such as employees, shifts, or shift_assignments.
4. Boolean-like values are stored as numbers: active = 1 means active/yes, active = 0 means inactive/no. is_manager works the same way.

12.Summary

The updated system supports persistent employee and shift management, driver and license handling, delivery validation through a mock delivery service, automatic restoration of initial data when the database is empty, and a controlled reset option for testing. The submitted database contains sample records to test employee management, availability, shift creation, manager assignment, role assignment, delivery validation, and database persistence.